

FORTYGUARD

Heat Intelligence Report

Heatmap Metadata

Location:	37.33279999945669, -121.88699999999997
Date and Time:	2024-10-02 12:00
Temperature:	4.5°C (40.1°F)



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Map



Figure 1: OverviewArea



Figure 2: Tile Area



Geographic Analysis

1. General Location

Identification

- **City:** San José, California, United States
- **Geographic classification:** Inland urban core situated in the southern portion of the Santa Clara Valley (commonly known as Silicon Valley); surrounded by foothills and mountains on three sides
- **Köppen climate zone:** Csb — Warm-summer Mediterranean climate, characterized by dry, warm summers and mild, wet winters. Precipitation is negligible from May through October.

Temperature Context

- **Provided temperature:** 40.74°C = **105.3°F**
- **Climate normals for October 2:** Average high temperatures for early October in San José typically range from 78–82°F, with an average daily mean near 68–70°F (based on 1991–2020 NOAA normals for San José International Airport, KSJC).
- **Departure from normal:** At 105.3°F, this reading is approximately **23–27°F above the normal high** for this date — an extreme anomaly. This would rank among the hottest October temperatures ever recorded in San José. The all-time October record high at KSJC is approximately 104°F (set in October 2020 during a late-season heat wave).
- **Notable heat patterns:** San José is susceptible to extreme heat events driven by offshore (Diablo) wind patterns, which compress and warm air descending from the interior. Late-season heat waves in September–October have become more frequent and intense, consistent with observed climate warming trends in California.
- A reading of 105.3°F in early October strongly suggests an active Diablo wind event or an exceptional ridge of high pressure, suppressing marine influence and driving extreme heating in the valley.

Sources:

- NOAA climate normals (1991–2020) for San José International Airport (KSJC)
- Köppen climate classification system
- NWS San Francisco Bay Area historical climate records

2. Terrain & Elevation

Elevation & Terrain Type

- **Elevation:** Approximately 80–100 feet above sea level in the central San José area near these coordinates.
- **Terrain type:** Broad, relatively flat alluvial valley floor (Santa Clara Valley), flanked by the Santa Cruz Mountains to the west/southwest (peaks to ~3,800 ft) and the Diablo Range/Hamilton Range to the east (peaks to ~4,200 ft).

Terrain Influence on Heat Dynamics

- **Valley basin effect:** The Santa Clara Valley acts as a semi-enclosed basin. Under high-pressure conditions, warm air pools in the valley with limited ventilation, amplifying daytime heating. Nocturnal cooling is also impaired because surrounding ridgelines reduce radiative heat loss and block regional-scale drainage flows.
- **Diablo wind dynamics:** During offshore wind events, air descends from the east over the Diablo Range, undergoing compressional (adiabatic) warming at roughly 5.5°F per 1,000 ft of descent. A descent of ~3,000–4,000 ft can add 16–22°F to air temperature, explaining extreme readings like 105.3°F.
- **Slope orientation:** The valley floor receives direct, unobstructed solar radiation throughout the day. West-facing slopes of the Diablo Range receive intense afternoon sun, contributing to thermal loading of air masses that settle into the valley.
- **Reduced marine influence:** The Santa Cruz Mountains form a significant barrier (~2,000–3,800 ft) to the cooling marine layer and fog from the Pacific. San José is notably warmer than coastal communities just 20 miles to the west due to this orographic blocking.

Sources:

- USGS National Elevation Dataset (NED)
- NWS understanding of Diablo wind meteorology



- General principles of adiabatic lapse rates and valley climatology

3. Proximity to Water Bodies

Nearest Significant Water Bodies

- **San Francisco Bay (South Bay/Lower Bay):** Approximately 8–10 miles to the northwest of central San José. The southern terminus of the Bay includes extensive salt ponds and marshlands near Alviso.
- **Pacific Ocean:** Approximately 20–25 miles to the west/southwest, separated by the Santa Cruz Mountains.
- **Local waterways:** The Guadalupe River and Coyote Creek flow through San José, but these are relatively narrow, channelized urban streams with minimal thermal moderating capacity.
- **Reservoirs:** Several reservoirs (Calero, Anderson, Coyote, Lexington) lie within 8–15 miles in the surrounding foothills, but their modest size limits regional thermal influence.

Thermal Moderating Effect

- **Bay influence:** Under normal conditions, onshore (northwesterly) breezes from the Bay provide modest cooling to San José, particularly in the afternoon. However, during offshore wind events (Diablo winds), this marine influence is completely negated — winds blow from the east/northeast, pushing warm continental air into the valley.
- **At 105.3°F in early October:** The marine layer is almost certainly absent. Offshore flow prevents any Bay-driven temperature buffering, which is the primary mechanism enabling such extreme temperatures.
- **Humidity:** Under normal conditions, proximity to the Bay maintains moderate humidity (40–60% in summer). During Diablo wind events, relative humidity can plummet to 10–20%, further reducing evaporative cooling and increasing sensible heat loading.
- **Seasonal variation:** Bay water temperatures in early October are near their annual peak (~60–65°F), which under normal wind regimes would provide modest evaporative cooling. This effect is irrelevant during offshore flow.

Sources:

- USGS hydrographic data for South San Francisco Bay
- NWS Bay Area wind climatology
- NOAA sea surface temperature records for San Francisco Bay

4. Green Spaces & Vegetation

Type and Extent of Vegetation

- **Urban parks:** San José has several significant parks including Guadalupe River Park (downtown linear park), Kelley Park (~156 acres), Alum Rock Park (~720 acres in the eastern foothills), and numerous neighborhood parks. Total city parkland exceeds 3,400 acres.
- **Street trees:** San José has an estimated 250,000+ street trees. The city's Community Forest Management Plan targets expanded canopy coverage.
- **Agriculture:** Historically the "Valley of Heart's Delight" for its orchards, the Santa Clara Valley has been largely urbanized. Some agricultural land remains on the southern and eastern periphery.
- **Estimated canopy coverage:** Approximately 15–18% citywide, which is below the national urban average of ~27%. Downtown and commercial areas have significantly lower canopy (~5–10%).

Seasonal and Cooling Impact

- **Vegetation type:** Mix of evergreen (live oaks, redwoods in riparian corridors, ornamental conifers) and deciduous species (sycamores, maples, fruit trees). In early October, deciduous canopy is still largely intact, providing near-peak shading.
- **Evapotranspiration:** By early October, after a dry summer with no rain since April/May, unirrigated vegetation and grasslands are dormant and brown. Only irrigated parks and street trees provide meaningful evapotranspiration. This significantly reduces the cooling potential of green spaces compared to spring.
- **Cooling contribution:** Well-irrigated parks in San José can be 5–10°F cooler than surrounding paved surfaces during peak heat. However, during extreme heat events with very low humidity and offshore winds, evapotranspiration efficiency drops, reducing this differential to an estimated 3–6°F.
- **Urban greening initiatives:** San José's "Our City Forest" nonprofit and the city's Green Vision plan have targeted planting 100,000+ trees. Cool-corridor and shade-equity programs focus on underserved neighborhoods with low canopy



coverage.

Sources:

- City of San José Community Forest Management Plan
- Our City Forest nonprofit data
- USDA Forest Service urban canopy assessments for California cities
- General urban forestry research on park cool island effects

5. Land Cover Type

Dominant Surface Materials

- **Central/Downtown San José:** Predominantly asphalt roadways, concrete sidewalks, and commercial rooftops. The downtown core has extensive impervious surface coverage estimated at 80–90%.
- **Suburban residential:** Mix of asphalt streets, concrete driveways, composite shingle roofing, and irrigated landscaping. Impervious fraction estimated at 50–65%.
- **Commercial/industrial zones:** North San José and areas near the airport feature large-footprint commercial buildings, expansive parking lots, and warehouse/data center campuses with very high impervious fractions (85–95%).
- **Bare soil/dry grassland:** Hillside open spaces to the east and south are covered in dry, golden grass by October — low albedo (~0.15–0.20) and minimal moisture content, contributing to sensible heat flux rather than cooling.

Thermal Properties

- **Asphalt:** Albedo ~0.05–0.10 (very low); surface temperatures can reach 150–170°F under 105°F ambient conditions. High thermal mass retains heat well into nighttime hours.
- **Concrete:** Albedo ~0.20–0.35; somewhat lower heat absorption than asphalt but still significant thermal storage.
- **Composite shingle roofs:** Albedo ~0.10–0.20 (dark colors); major contributors to building heat gain and re-radiation.
- **Cool-roof and cool-pavement programs:** California's Title 24 building energy code requires cool roofing (solar reflectance index ≥ 0.20 for steep-slope, ≥ 0.63 for low-slope) on new commercial and some residential construction. San José has adopted these standards. Some tech campuses in North San José feature high-albedo roofing and green roofs, but adoption remains limited across the broader urban area.

Sources:

- California Energy Commission Title 24 Building Standards
- NLCD (National Land Cover Database) impervious surface data
- General thermal properties of urban surface materials from EPA Heat Island literature

6. Building Density & Height

Building Density Characterization

- **Downtown San José:** Mid-rise to high-rise core. Recent development has added numerous residential towers (15–25 stories) and office buildings (10–20 stories). The tallest structures reach approximately 280–300 feet. Building density is moderate-to-high but less intense than San Francisco's downtown.
- **North San José / Silicon Valley corridor:** Low-to-mid-rise commercial and tech campus development (2–6 stories), with large building footprints and extensive surface parking. This morphology creates broad, exposed heat-absorbing surfaces with minimal shading between structures.
- **Residential neighborhoods:** Predominantly single-family, 1–2 story homes with moderate lot coverage. Older neighborhoods (e.g., Willow Glen, Rose Garden) have more mature tree canopy; newer developments have less.

UHI Contribution from Built Form

- **Vertical massing:** Downtown's growing cluster of tall buildings creates localized heat trapping through multiple reflections of shortwave radiation and reduced sky view for longwave radiation escape. However, the relatively compact downtown footprint limits this effect to a small area.
- **Sprawling commercial zones:** The dominant UHI contributor in San José is the extensive low-rise commercial/industrial fabric with dark roofs and parking lots, which absorbs enormous amounts of solar radiation across a wide area.
- **Anthropogenic heat:** Significant sources include vehicle traffic (San José has very high car dependency; major



freeways I-280, I-680, US-101, and CA-87 carry heavy traffic), HVAC systems operating at maximum capacity during a 105°F event (estimated to add 1–3°F to local air temperatures in dense commercial areas), and data centers in North San José which generate substantial waste heat year-round.

- **UHI magnitude:** San José's UHI intensity is estimated at 4–8°F during typical summer evenings relative to surrounding rural/agricultural areas. During an extreme heat event like this, the nocturnal UHI could reach 8–12°F due to massive thermal storage in impervious surfaces.

Sources:

- City of San José General Plan and Urban Village planning documents
- EPA Urban Heat Island research for mid-size U.S. cities
- Typical urban morphology for Silicon Valley cities

7. Shadow Coverage

Solar Angle Calculation

- **Date/Time:** October 2, 2024, at 12:00 hours (solar noon is approximately 12:50 PM PDT at this longitude, so 12:00 PM is slightly before solar noon).
- **Solar declination:** Approximately -4.0° on October 2.
- **Solar elevation at noon:** Estimated at approximately $53\text{--}55^\circ$ above the horizon (calculated as $90^\circ - \text{latitude} + \text{declination} \approx 90^\circ - 37.29^\circ + (-4.0^\circ) \approx 48.7^\circ$ at solar noon; at 12:00 local, slightly lower due to being ~50 minutes before solar noon — estimated $\sim 49\text{--}52^\circ$).
- **Solar azimuth:** Approximately $170\text{--}175^\circ$ (nearly due south, slightly east of south at 12:00 PM).

Shadow Coverage Assessment

- **Building shadows:** At a solar elevation of $\sim 50^\circ$, a 100-foot building casts a shadow approximately 84 feet long to the north. In downtown San José, taller buildings (200–300 ft) cast shadows of 170–250 ft, providing meaningful shade on north-facing sidewalks and adjacent lower structures. However, shadow coverage in the downtown core is still limited at midday due to the relatively high sun angle.
- **Suburban areas:** Single-story (12–15 ft) residential buildings cast shadows of only 10–13 ft — negligible for cooling adjacent outdoor spaces at midday.
- **Tree shadows:** Mature street trees (30–50 ft canopy height) with full October canopy provide shade patches of approximately 25–40 ft diameter. Where present, tree shade can reduce surface temperatures by 20–40°F and ambient air temperatures by 3–7°F compared to fully exposed pavement.
- **Permanent vs. seasonal shade:** Building shade is year-round but varies with solar angle. Deciduous tree shade is still available in early October (leaf drop typically occurs November–December in San José). Evergreen species provide consistent shade year-round.
- **Overall assessment:** At midday on October 2, shadow coverage across San José is minimal — estimated at 10–15% of ground area in the downtown core and 5–8% in suburban/commercial areas. This leaves the vast majority of surfaces exposed to intense direct solar radiation, contributing directly to the extreme temperature reading.

Sources:

- Solar position calculations based on NOAA Solar Calculator methodology
- General urban shading research from Lawrence Berkeley National Laboratory
- EPA guidance on shade and surface temperature reduction

8. Urban Geometry & Street Canyon Effects

Street Canyon Characteristics

- **Downtown San José:** The downtown street grid features blocks approximately 300–400 ft wide with streets approximately 60–80 ft wide (curb to curb plus sidewalks). Building heights range from 40–300 ft, yielding height-to-width (H/W) ratios of approximately 0.5–4.0, with most blocks in the 0.7–2.0 range. Higher H/W canyons (>1.5) trap longwave radiation at night, slowing nocturnal cooling by an estimated 3–6°F compared to open areas.
- **Suburban/commercial areas:** H/W ratios are very low (0.1–0.3) due to single-story buildings set back from wide arterial roads (often 100+ ft wide including parking setbacks). These open geometries allow maximum solar access during the day but also permit more efficient radiative cooling at night.



Wind Corridors and Ventilation

- **Street layout:** Downtown San José follows a regular grid oriented approximately NW-SE and NE-SW. Under normal conditions, the prevailing NW afternoon breeze from the Bay aligns reasonably well with NW-SE streets, allowing some ventilation. However, during the Diablo wind event implied by 105.3°F, winds are from the east/northeast — these align poorly with the grid, reducing canyon ventilation.
- **Wind corridor presence:** The Guadalupe River corridor provides a modest linear ventilation pathway through the urban core, but it is narrow and flanked by development. Major freeway corridors (US-101, I-280) act as open channels but generate their own anthropogenic heat from traffic.
- **Convective cooling impact:** During offshore wind events, the hot, dry east winds actually increase temperatures rather than providing cooling. Wind speeds during Diablo events can reach 20–40 mph in exposed areas but are reduced to 5–15 mph within the urban canopy layer, limiting even the modest convective heat removal that wind provides.

Sky View Factor (SVF)

- **Downtown core:** Estimated SVF of 0.3–0.5 in the densest canyon sections, reducing to 0.2–0.3 in the tallest building clusters. Lower SVF impedes nocturnal longwave radiation escape, contributing to elevated nighttime temperatures — a critical health concern during multi-day heat events.
- **Suburban residential:** SVF estimated at 0.7–0.9 (high openness), allowing efficient nighttime cooling but also maximum daytime solar exposure.
- **Commercial/tech campus:** SVF of 0.6–0.8, with large open parking areas having SVF approaching 1.0 — these areas cool relatively efficiently at night but experience extreme daytime surface heating.
- **Net effect:** San José's overall urban geometry is dominated by low-rise, open morphology, which means the primary heat challenge is massive daytime solar absorption across expansive impervious surfaces rather than nocturnal trapping in deep canyons. The downtown canyon effect is a secondary but growing concern as high-rise development intensifies.

Sources:

- Urban canyon radiation trapping research (Oke, 1987 — boundary layer climatology principles)
- City of San José street design standards
- General SVF estimation methods for North American cities of similar morphology
- NWS Bay Area Diablo wind characterization

Environmental Factors Analysis

1. Air Quality & Atmospheric Opacity

- The coordinates 37.29°N, 121.83°W correspond to San Jose, California, in the Santa Clara Valley (Silicon Valley). Note: The input lists "121.83 E" but contextually this is clearly 121.83°W given the latitude pairing.
- Typical AQI for early October in San Jose ranges from 30–80 (Good to Moderate per EPA classification), though wildfire season can push AQI into Unhealthy for Sensitive Groups (101–150) or higher.
- Primary pollutants of concern: Ground-level ozone (O₃) is the dominant summer/early fall pollutant due to photochemical reactions; PM_{2.5} is elevated during wildfire events and temperature inversions; NO₂ from vehicle traffic on major corridors (I-680, I-280, US-101).
- Aerosol radiative forcing: Elevated PM_{2.5} from wildfires can reduce incoming solar radiation by 10–30%, temporarily lowering daytime highs, but also traps outgoing longwave radiation at night, suppressing nocturnal cooling. Under clear conditions, the relatively clean air allows high solar transmittance, amplifying surface heating.
- Seasonal factors: October marks the tail end of California's fire season; Diablo wind events (offshore, dry, hot winds) can simultaneously degrade air quality and raise temperatures dramatically.
- Feedback loop: At 105.3°F, intense heat accelerates photochemical ozone production (VOCs + NO_x + UV → O₃). Higher temperatures also increase biogenic VOC emissions from vegetation, further fueling ozone formation. Stagnant high-pressure systems trap pollutants in the valley.

Sources:

- EPA Air Quality Index (AQI) standards and breakpoints
- Bay Area Air Quality Management District (BAAQMD) monitoring data
- EPA photochemical ozone formation science

2. Climate Classification & Historical Weather



- Köppen classification: **Csb** (Warm-summer Mediterranean climate) — characterized by dry, warm summers and mild, wet winters. Some lower-elevation portions of San Jose approach Csa (Hot-summer Mediterranean).
- Historical climate normals for San Jose (October 2): Typical high ~77°F (25°C), typical low ~54°F (12°C), mean ~65°F (18.3°C). These are based on the 1991–2020 climate normals period.
- The provided temperature of 40.74°C converts to **105.3°F**. This is approximately **28°F above the normal high** for this date — an extraordinary departure representing an extreme heat event.
- Extreme temperature records: San Jose's all-time record high is 109°F (June 2000). October record highs reach approximately 100–104°F. A reading of 105.3°F in early October would approach or exceed the monthly record, consistent with the type of extreme heat event driven by offshore (Diablo) wind patterns.
- The Mediterranean climate's dry fall season means minimal cloud cover and low soil moisture, which eliminates evaporative cooling and allows extreme temperature excursions during heat events. The valley geography (ringed by the Diablo Range to the east and Santa Cruz Mountains to the west) can trap heat under subsidence inversions.

Sources:

- Köppen climate classification system
- NOAA/NWS climate normals (1991–2020) for San Jose, CA (Station KSJC)
- NOAA historical extreme temperature records for Santa Clara County

3. Climate Trends & Projections

- Observed warming trend in the San Jose/Bay Area region: approximately 2.0–3.0°F since 1970, with warming accelerating to roughly 0.5–0.7°F per decade in recent decades. Nighttime minimum temperatures have warmed faster than daytime highs.
- Heatwave frequency has increased significantly: extreme heat events (defined as days exceeding 100°F) in inland Bay Area locations have roughly doubled in frequency since the 1980s.
- Urban expansion: Santa Clara Valley has experienced massive urbanization since the 1960s (population growth from ~600,000 to ~2 million in the county), converting orchards and agricultural land to impervious surfaces, intensifying UHI effects.
- Climate projections (RCP 4.5/SSP2-4.5): By the 2030s, average temperatures projected to increase 1.5–2.5°F above the 1991–2020 baseline; by 2050s, 3–5°F increase. Extreme heat days (>100°F) projected to increase 2–4x by mid-century.
- Cooling Degree Days (CDD, base 65°F): Increasing trend of approximately 50–100 CDD per decade in the South Bay reflecting growing cooling energy demand and cumulative heat exposure.
- The occurrence of a 105°F reading in early October is consistent with the trend toward later-season extreme heat events extending the "heat season" beyond its historical bounds.

Sources:

- Cal-Adapt climate projections (California's Fourth Climate Change Assessment)
- NOAA climate trend analyses for California climate divisions
- U.S. Census Bureau population data for Santa Clara County
- IPCC AR6 regional projections for western North America

4. Soil Moisture & Permeability

- Dominant soil types in the Santa Clara Valley floor: clay loams and silty clay loams (alluvial deposits from surrounding mountains). These soils have moderate-to-low permeability when dry and can become nearly impermeable when compacted or sealed.
- Expected soil moisture in early October: **Extremely low**. California's Mediterranean climate produces virtually no rainfall from May through October. By early October, soils are at or near their seasonal minimum moisture content (volumetric water content typically 5–15% in upper layers).
- Heat implications: Dry clay soils have high thermal conductivity but minimal evaporative cooling potential. Without moisture, latent heat flux is near zero, meaning nearly all absorbed solar energy converts to sensible heat, raising surface and air temperatures. Dry bare soil surface temperatures can reach 140–160°F under peak solar load.
- Urbanization impact: Approximately 60–70% of the San Jose urban core is covered by impervious surfaces (buildings, roads, parking lots), eliminating natural soil moisture pathways and concentrating runoff rather than infiltration. This dramatically reduces evapotranspiration capacity.
- Groundwater: The Santa Clara Valley groundwater basin has been managed to prevent overdraft (historically caused subsidence), but the water table is too deep (typically 20–100+ feet) to contribute meaningfully to surface moisture availability during the dry season.

Sources:



- USDA Natural Resources Conservation Service (NRCS) soil surveys for Santa Clara County
- Santa Clara Valley Water District groundwater monitoring
- California Irrigation Management Information System (CIMIS) soil moisture data

5. Heat Retention Capacity of Surfaces

- Dominant surface materials in San Jose: Asphalt roadways and parking lots (~30–35% of urban area), concrete sidewalks and structures (~15–20%), roofing materials (mixed — dark shingles, flat commercial roofs, some cool roofs), landscaped vegetation (~20–25%), and remaining mixed surfaces.
- Surface temperature differentials under peak solar load at 105°F air temperature:
 - Black asphalt: 150–170°F (60–75°F above air temperature)
 - Concrete: 130–150°F (25–45°F above air temperature)
 - Dark roofing: 155–175°F
 - Irrigated grass: 85–95°F (10–20°F below air temperature due to evapotranspiration)
 - Bare dry soil: 135–155°F
- Thermal properties: Asphalt albedo ~0.05–0.15 (absorbs 85–95% of solar radiation), thermal admittance ~1,500–2,000 J/(m²·K·s^{0.5}). Concrete albedo ~0.20–0.35, thermal admittance ~1,500–2,500 J/(m²·K·s^{0.5}). Both materials have high volumetric heat capacity, storing energy during the day and releasing it at night.
- UHI contribution: The dominance of dark, low-albedo surfaces in commercial and industrial zones (North San Jose, along I-880 corridor) creates intense localized heat islands. Estimated UHI intensity contribution from surface materials alone: 3–7°F above surrounding vegetated areas.
- Cooling infrastructure: San Jose has adopted cool roof requirements in Title 24 building code updates; some commercial developments include reflective pavements. The city's urban forest canopy covers approximately 15% of the city, below the national urban average of ~27%.

Sources:

- EPA Heat Island Effect documentation and surface temperature studies
- Lawrence Berkeley National Laboratory (LBNL) cool surfaces research
- California Title 24 Building Energy Efficiency Standards
- City of San Jose Urban Forest Master Plan

6. Humidity Levels

- Typical relative humidity for San Jose at noon in early October: 30–45% under normal conditions. However, during an extreme heat event driven by offshore (Diablo/Santa Ana-type) winds, relative humidity drops dramatically to **10–20%** or even single digits.
- Estimated dew point during this event: 35–50°F (2–10°C). The extremely low dew point indicates continental or subsiding air mass, characteristic of offshore flow patterns. This is well below the dangerous heat stress threshold of 75°F dew point.
- Heat index implications: At 105.3°F with 10–20% RH, the heat index is actually close to or slightly below the actual air temperature (~103–106°F), because low humidity allows effective evaporative cooling from skin. However, this does NOT mean conditions are safe — direct solar radiation and extreme dry heat still pose serious risks including rapid dehydration.
- Seasonal humidity pattern: October in San Jose is typically the driest month before fall rains begin (usually late October–November). The dry season suppresses latent heat flux, meaning more energy goes into sensible heating of the air.
- Nocturnal cooling implications: Low humidity is actually favorable for nighttime radiative cooling (dry air is more transparent to longwave radiation). However, the extreme thermal mass of urban surfaces can override this advantage. During Diablo wind events, warm offshore winds may prevent cooling regardless of humidity levels.

Sources:

- NOAA/NWS heat index calculations and charts
- NOAA climate normals for humidity at KSJC
- NWS San Francisco Bay Area forecast office — Diablo wind event documentation

7. Solar Radiation Intensity

- Expected Global Horizontal Irradiance (GHI) at solar noon on October 2 at 37.29°N: approximately **750–850 W/m²** under clear skies. During an extreme heat event (typically associated with high pressure and cloudless skies), GHI would be at the upper end of this range.
- Solar elevation angle at solar noon on October 2 at 37.29°N: approximately **52–54°** (calculated from declination of ~-4° on



Oct 2). This is moderately high, producing intense direct beam radiation though somewhat less than summer solstice (~76°).

- Peak sunlight hours: Solar noon occurs around 12:50 PM PDT in San Jose. Peak air temperature typically lags solar noon by 2–3 hours (occurring around 2:00–4:00 PM), meaning at 12:00 hours the temperature may still be rising toward its daily maximum — the actual peak could exceed 105.3°F.
- Terrain and urban modification: The Santa Clara Valley floor receives largely unobstructed solar radiation. Urban canyon geometry in downtown San Jose (moderate building heights, 3–15 stories) creates some shading on east-west streets but north-south oriented streets receive full midday exposure. The surrounding mountains provide no meaningful shading at midday.
- Differential absorption under this solar load: At ~800 W/m², asphalt absorbs ~680–760 W/m² (albedo 0.05–0.15), while concrete absorbs ~520–640 W/m² (albedo 0.20–0.35). Irrigated vegetation absorbs similar total radiation but converts 50–70% to latent heat through evapotranspiration, dramatically reducing sensible heat contribution.

Sources:

- National Solar Radiation Database (NSRDB) — NREL
- Solar position algorithms (NOAA Solar Calculator)
- ASHRAE solar radiation fundamentals

8. Nighttime Cooling Rates

- Typical Diurnal Temperature Range (DTR) for San Jose in early October **25–30°F** under normal conditions (high ~77°F, low ~52°F). During extreme heat events, DTR may compress to 20–25°F if warm offshore winds persist overnight, or may remain large (30–35°F) if winds subside.
- Urban thermal mass effect: The extensive asphalt and concrete surfaces in San Jose store enormous quantities of heat during a 105°F day. These surfaces remain elevated (90–110°F surface temperature) well into the evening, releasing stored energy as longwave radiation and convective heat, keeping air temperatures elevated.
- Nighttime cooling suppression factors:
 - Thermal mass of urban surfaces: Primary factor — releases heat for 6–10 hours after sunset
 - Urban canyon geometry: Reduced sky view factor in denser areas traps outgoing radiation
 - Anthropogenic heat: HVAC systems operating at maximum capacity reject substantial waste heat (estimated 5–15 W/m² in commercial districts)
 - Warm offshore winds (if Diablo event): Can maintain temperatures in the 80s–90s°F overnight
- Estimated nighttime low during this event: 70–80°F (compared to normal of ~54°F). Consecutive hot nights above 75°F are particularly dangerous as the body cannot recover from daytime heat stress.
- Urban-rural nighttime differential: Estimated 5–10°F warmer in urban San Jose compared to surrounding rural/agricultural areas during this event, with the greatest differential in commercial/industrial zones along the I-880 corridor.

Sources:

- NOAA climate normals for diurnal temperature range
- EPA Urban Heat Island studies — nighttime UHI intensity
- Oke, T.R. — boundary layer climatology principles (urban energy balance)

9. Thermal Comfort Index

- Most applicable metric: **Universal Thermal Climate Index (UTCI)** is most appropriate here because it integrates radiation load, wind speed, humidity, and air temperature — all critical during an extreme dry heat event where radiant heat dominates but humidity is low. The Heat Index alone would underestimate danger by not accounting for solar radiation exposure.
- Estimated conditions at 12:00 hours:
 - Air temperature: 105.3°F (40.7°C)
 - Relative humidity: ~10–20%
 - Wind speed: Estimated 5–15 mph (Diablo winds may be stronger in exposed areas)
 - Mean radiant temperature: Estimated 140–160°F (60–70°C) in full sun due to reflected radiation from pavement and direct solar beam
- Estimated UTCI: **42–46°C (108–115°F equivalent)**, corresponding to **"Very Strong Heat Stress"** category (UTCI 38–46°C). In full sun near dark pavement, conditions may approach **"Extreme Heat Stress"** (>46°C UTCI).
- Vulnerable populations at severe risk:
 - Outdoor workers (construction, agriculture, landscaping): Risk of heat stroke within 1–2 hours without adequate rest/hydration
 - Elderly and those with cardiovascular conditions: Thermoregulatory failure risk



- Homeless population (significant in San Jose): No access to cooling, chronic dehydration
- Children: Higher metabolic heat production relative to body surface area
- Local mitigation: San Jose activates cooling centers during extreme heat events; some transit stops have shade structures; limited misting infrastructure exists. The city's relatively low tree canopy (15%) limits pedestrian shade availability.

Sources:

- UTCI assessment scale (ISB Commission 6)
- NIOSH heat stress guidelines for occupational exposure
- City of San Jose Emergency Services — extreme heat protocols
- NWS Excessive Heat Warning criteria

10. Seasonal Variability of Urban Heat Islands

- UHI intensity variation by season in San Jose:
 - **Summer/Early Fall (June–October):** Maximum UHI intensity of 5–9°F, driven by high solar input, dry conditions eliminating rural evaporative cooling advantage, and peak HVAC waste heat. This is the period of greatest UHI intensity.
 - **Late Fall/Winter (November–February):** Reduced UHI intensity of 3–5°F, as cloud cover and rain reduce differential heating, and rural areas retain more moisture.
 - **Spring (March–May):** Moderate UHI intensity of 4–6°F, increasing as soils dry and vegetation transitions.
- Driving factors for seasonal variation:
 - Foliage: Deciduous trees (limited in San Jose's largely evergreen urban forest) provide slightly less shade in winter but this is a minor factor
 - Soil moisture differential: The strongest driver — rural areas dry out in summer, reducing their evaporative cooling advantage and partially converging with urban temperatures. However, irrigated urban landscaping can create "oasis effects" locally.
 - Energy use: Summer cooling loads generate substantial anthropogenic heat; winter heating loads are lower in this mild climate
 - Wind patterns: Summer sea breeze from San Francisco Bay moderates coastal-facing areas but weakens inland; Diablo winds in fall eliminate this cooling entirely
- Urban-rural differential during this extreme event: Estimated **7–12°F** warmer in urban core compared to nearby rural/open space areas (e.g., Coyote Valley to the south), representing near-maximum UHI expression.
- Greatest compound heat risk season: **Late September through mid-October** — this period combines residual summer heat storage in urban materials, maximum soil dryness, potential for Diablo wind events, and reduced day length that concentrates solar energy. The combination of extreme temperatures with degraded air quality (wildfire smoke) creates compound health risks exceeding those of mid-summer.
- Adaptation measures: San Jose's Climate Smart plan includes urban tree planting targets, cool roof mandates, green infrastructure requirements for new development, and the preservation of Coyote Valley as a cooling corridor. Implementation remains ongoing.

Sources:

- EPA Urban Heat Island assessment methodologies
- City of San Jose Climate Smart Plan (2018)
- NASA/USGS Landsat thermal imagery studies of Bay Area UHI
- California Office of Environmental Health Hazard Assessment — compound heat-air quality risk assessments

Urban Factors Analysis

1. Urban Land Use Characteristics

Location Context

- The coordinates (37.29°N, 121.83°W) place this location in the heart of **San José, California**, within the southern portion of the San Francisco Bay Area — specifically near the central/downtown corridor of the city.
- The average temperature of **40.74°C (105.3°F)** on October 2, 2024 represents an extreme heat event well above San José's typical early-October highs (~78–82°F), indicating a significant heat wave condition that amplifies all urban heat drivers discussed below.



Dominant Land Use Composition

- **Residential (~45–50%):** A mix of single-family residential neighborhoods (particularly to the south and east) and increasing medium-density multifamily housing near downtown and transit corridors. Many neighborhoods feature older ranch-style homes with modest lot sizes.
- **Commercial/Mixed-Use (~20–25%):** Downtown San José contains office towers, retail, and mixed-use developments. Major commercial corridors run along key arterials (e.g., Stevens Creek Boulevard area, Santana Row district to the west).
- **Industrial/Light Industrial (~10–15%):** North San José contains a significant concentration of technology campuses, light manufacturing, and data center facilities. The area near the Guadalupe River corridor and north toward Milpitas has industrial legacy zones.
- **Institutional/Transportation (~10%):** San José Mineta International Airport lies approximately 2 miles to the northwest, along with associated logistics and transportation infrastructure. San José State University campus is near downtown.
- **Open Space/Parks (~5–8%):** Scattered urban parks, riparian corridors along the Guadalupe River and Coyote Creek, and regional open space at the urban fringe.

Impervious Surface Fraction

- **Estimated impervious surface fraction: 65–75%** in the central/downtown area, decreasing to 50–60% in suburban residential zones. This is consistent with NLCD (National Land Cover Database) classifications for urbanized areas in Santa Clara County.
- Extensive surface parking lots associated with commercial centers and older retail corridors contribute disproportionately to impervious coverage relative to their economic productivity.
- At 105.3°F, these impervious surfaces are likely reaching surface temperatures of 140–165°F, creating intense sensible heat flux into the urban boundary layer.

Heat Differentials from Land Use Patterns

- High-density commercial zones downtown and tech campuses in North San José create concentrated heat islands, with measured UHI intensities typically 5–9°F above surrounding suburban areas during extreme heat events.
- Green corridors along the Guadalupe River provide linear cooling features but are narrow and do not penetrate deeply into adjacent built-up neighborhoods.
- The segregation of industrial/commercial heat-generating zones in North San José from residential areas to the south and east creates a thermal gradient, but prevailing northwesterly winds can advect industrial waste heat into residential zones.

Zoning and Planning Policies

- San José's **Envision San José 2040 General Plan** promotes urban village concepts with higher-density, mixed-use development along transit corridors — which can intensify localized heat if not paired with green infrastructure.
- The city has adopted a **Green Stormwater Infrastructure Plan** that encourages permeable surfaces and bioretention, which has co-benefits for urban cooling.
- San José's **Climate Smart San José** initiative includes goals for urban canopy expansion, though implementation has lagged behind targets.

Thermal Equity Implications

- Heat-mitigating green infrastructure is unevenly distributed: wealthier neighborhoods in west San José (e.g., Willow Glen, Rose Garden) tend to have more mature tree canopy, while lower-income neighborhoods in east San José have significantly less canopy cover and more impervious surface — a well-documented environmental justice concern in this city.
- At 105.3°F, this inequity becomes life-threatening, as heat-vulnerable populations in under-canopied neighborhoods face substantially higher exposure.

Sources:

- USGS National Land Cover Database (NLCD) for Santa Clara County impervious surface estimates
- City of San José Envision 2040 General Plan
- Climate Smart San José plan documentation
- Published urban heat island studies for San José / Santa Clara Valley (e.g., research from San José State University Department of Meteorology and Climate Science)



2. Anthropogenic Heat Sources

Major Economic Activity and Waste Heat

- San José is the economic hub of **Silicon Valley**, and the dominant economic activities — technology, software, hardware R&D, and data processing — generate substantial waste heat through data centers, server farms, and large commercial office complexes.
- **Data centers** are a critical anthropogenic heat source. North San José and adjacent areas host numerous data center facilities for major technology companies. These facilities reject enormous quantities of heat: a typical 20 MW data center rejects approximately 60–70 million BTU/hour of waste heat to the atmosphere via cooling towers and HVAC systems. Estimated: North San José contains cumulative data center capacity in the range of 100–200+ MW, contributing a localized thermal footprint measurable at the neighborhood scale.
- At 105.3°F ambient temperature, data center and commercial HVAC cooling efficiency drops significantly (COP decreases by ~2–4% per °F above design conditions), meaning **more electricity is consumed and more waste heat is rejected** per unit of cooling — a positive feedback loop that intensifies outdoor heat.

Transportation Heat

- **Highway proximity:** Interstate 280 runs through the western portion of the city, US-101 runs through the eastern/northern portion, and Interstate 880 borders the north. Highway 87 (Guadalupe Expressway) bisects the city north-south near downtown. These freeways carry combined average daily traffic volumes exceeding 500,000 vehicles in the metro area.
- Vehicle exhaust and engine waste heat contribute an estimated 30–50 W/m² of anthropogenic heat flux along major highway corridors during peak traffic periods.
- San José Mineta International Airport (~2 miles NW) contributes jet exhaust heat and extensive impervious tarmac surfaces that radiate stored heat, particularly during afternoon hours at 105.3°F.

HVAC Rejection Heat

- At 105.3°F, virtually every commercial and residential building is running air conditioning at or near maximum capacity. For a city of ~1 million people with approximately 350,000 housing units and tens of millions of square feet of commercial space, aggregate HVAC waste heat rejection is enormous.
- Estimated: Commercial HVAC systems reject approximately 1.2–1.4 units of heat outdoors for every unit of cooling delivered indoors. At extreme temperatures, this ratio worsens. The cumulative effect raises ambient air temperature in dense commercial districts by an estimated 1–3°F above what would occur without HVAC operation.
- Rooftop condensing units on commercial buildings create localized hot spots, particularly in low-rise commercial strips where units are at pedestrian-accessible heights.

Impervious Commercial Surfaces

- Large surface parking lots associated with shopping centers, auto dealerships (San José has extensive auto row corridors), and big-box retail zones act as thermal batteries. At 105.3°F ambient, asphalt parking lot surface temperatures can reach 150–170°F.
- Typical for this area: Retail corridors along Stevens Creek Boulevard, Tully Road, and Story Road feature extensive surface parking with minimal shade tree coverage, creating some of the hottest microenvironments in the city.

Sustainability Initiatives

- San José has one of the highest per-capita solar installation rates among major US cities, with the **San José Clean Energy** community choice energy program providing high-renewable-content electricity. Solar panels on rooftops have a modest cooling co-benefit by shading the roof surface beneath them (reducing roof surface temperature by ~5–10°F).
- The city has adopted California's Title 24 energy code, which mandates increasingly stringent building energy efficiency standards, including cool roof requirements for new commercial construction.
- LEED-certified buildings are common in newer tech campus developments in North San José and downtown, though they represent a minority of total building stock.
- No district cooling system is known to be operational at scale in San José, though some campus-level chilled water systems exist at technology company headquarters.

Notable Heat-Contributing Zones

- **North San José tech/industrial corridor:** ~3–5 miles north of the coordinates; high concentration of data centers, office



parks with large parking structures, and logistics facilities. Significant waste heat contributor.

- **San José Mineta International Airport:** ~2 miles NW; extensive impervious tarmac, jet operations, and vehicle traffic generate concentrated heat.
- **Downtown San José:** Immediately surrounding the coordinates; dense commercial activity, HVAC loads, vehicle traffic, and limited vegetation create a pronounced local heat island.

Sources:

- US DOE data center energy consumption estimates
- Caltrans traffic volume data for Santa Clara County highways
- California Energy Commission Title 24 building standards
- San José Clean Energy program documentation
- Published anthropogenic heat flux studies for US metropolitan areas (Sailor & Lu, 2004; related literature)

3. Green Infrastructure & Urban Cooling Systems

Urban Vegetation Strategy

- San José has an estimated **urban tree canopy coverage of approximately 15–18%**, which is below the national average for cities of comparable size (~25–30%) and significantly below the city's own stated goal of increasing canopy cover.
- The city's **Community Forest Management Plan** and the nonprofit **Our City Forest** have conducted tree planting campaigns, but canopy growth is slow and faces challenges from drought, development pressure, and maintenance funding.
- At 105.3°F, the existing canopy is critically important but insufficient to meaningfully reduce city-wide temperatures. Mature trees can reduce air temperature beneath them by 5–10°F through shading and evapotranspiration, but coverage gaps leave most pedestrian areas exposed.

Cooling Mechanisms

- **Evapotranspiration:** During extreme heat with adequate soil moisture, a mature urban tree can transpire 40–100 gallons of water per day, providing a cooling effect equivalent to ~230,000 BTU. However, October in San José follows the dry season, and soil moisture is typically depleted, reducing evapotranspiration rates by an estimated 30–50% compared to spring conditions.
- **Shading:** Where present, tree canopy reduces ground-level solar radiation by 60–90%, lowering surface temperatures of shaded pavement by 20–40°F compared to sunlit surfaces. This translates to an air temperature reduction of approximately 3–7°F at pedestrian level in well-shaded areas.
- Estimated: At 105.3°F, even well-shaded areas under mature canopy would still experience air temperatures of ~98–102°F — still above heat stress thresholds for vulnerable populations.

Corporate and Public Greening

- Major technology campuses in North San José and other parts of Silicon Valley (e.g., Adobe's downtown headquarters) incorporate landscaped grounds and some green roof elements, but these are localized and do not significantly influence neighborhood-scale temperatures.
- Rooftop gardens and vertical greening systems remain uncommon in San José compared to cities with stronger mandates (e.g., Singapore, Portland). Adoption is primarily voluntary and limited to high-profile developments.

Engineered Cooling Features

- San José does not have large-scale engineered urban cooling systems such as district cooling networks or extensive fountain/misting infrastructure in public spaces.
- Some public plazas downtown (e.g., near the San José Convention Center and City Hall area) include water features, but their thermal impact radius is limited to approximately 50–100 feet and provides minimal relief at 105.3°F.
- Lake Cunningham Park (~5 miles east) and other park water features provide localized cooling but are too distant to influence conditions at the coordinates.

Seasonal Variability

- San José's Mediterranean climate means that by early October, deciduous trees retain most of their foliage, so shading capacity is still near peak. However, water stress from the preceding dry summer significantly reduces evapotranspiration



effectiveness.

- Irrigated landscapes (parks, corporate campuses) maintain higher cooling capacity than unirrigated areas, but water conservation mandates during drought years can reduce irrigation and thus cooling capacity.
- **Assessment:** At 105.3°F, existing green infrastructure is **grossly inadequate** to maintain thermal comfort. The combination of insufficient canopy coverage and reduced evapotranspiration from dry conditions means green infrastructure provides only marginal mitigation during this extreme event.

Cooling Equity

- Tree canopy distribution in San José is strongly correlated with income and historical redlining patterns. Neighborhoods in **east San José** (e.g., Mayfair, East Side) have canopy coverage as low as 8–12%, while wealthier neighborhoods in **west San José** (e.g., Willow Glen, Cambrian) have canopy coverage of 20–30%.
- This disparity means that during a 105.3°F heat event, low-income communities of color experience disproportionately higher heat exposure — a well-documented environmental justice issue in San José.

Notable Cooling Infrastructure

- **Guadalupe River Park and Gardens:** A linear park running north-south through downtown, ~0.5 miles west of the coordinates. Provides a narrow green corridor with riparian vegetation but limited canopy width.
- **St. James Park:** A historic downtown park ~0.3 miles from the coordinates; provides localized shade but is small (~7 acres).
- **Kelley Park / Happy Hollow:** ~3 miles southeast; a larger park with mature trees providing more substantial cooling, but too distant to influence downtown conditions.
- **Plaza de César Chávez:** A small downtown park near the coordinates; provides modest shade and is a key public gathering space but has limited cooling capacity at extreme temperatures.

Sources:

- City of San José Community Forest Management Plan
- Our City Forest nonprofit program data
- USDA Forest Service i-Tree urban canopy assessments for San José
- Published research on urban tree canopy equity in the San Francisco Bay Area (e.g., studies from UC Berkeley and San José State University)
- EPA urban heat island mitigation literature on evapotranspiration cooling effects

4. Building Material Thermal Properties

Dominant Building Materials

- **Commercial/office buildings (downtown):** Primarily steel-frame construction with glass curtain walls, concrete cores, and flat roofs with built-up or single-ply membrane roofing. Newer tech office buildings feature extensive glazing.
- **Residential (suburban):** Wood-frame construction with stucco exterior (very common in San José), asphalt shingle roofing, and concrete slab-on-grade foundations. Older homes (1950s–1970s) have minimal insulation.
- **Industrial/commercial (North San José):** Tilt-up concrete panel construction with metal roofing — a dominant building type in Silicon Valley's industrial/tech zones. These structures have high thermal mass.
- **Paving:** Asphalt dominates roadways and parking lots; concrete sidewalks and some concrete roadways in newer developments.

Albedo Values

- **Fresh asphalt:** 0.04–0.05 (extremely low; absorbs ~95% of solar radiation)
- **Aged/weathered asphalt:** 0.10–0.15
- **Concrete sidewalks/roads:** 0.20–0.35
- **Stucco exterior walls (light-colored):** 0.30–0.45
- **Standard built-up roofing (dark):** 0.05–0.15
- **Cool roof coatings (white/reflective):** 0.60–0.70
- **Glass curtain walls:** 0.05–0.10 (absorptive) to 0.20–0.35 (reflective coatings), but reflected solar energy is redirected to streets and adjacent surfaces rather than being returned to the sky
- **Tilt-up concrete panels (light gray):** 0.20–0.30



- At 105.3°F with intense solar radiation (~900–1000 W/m² at solar noon in early October at this latitude), low-albedo surfaces absorb enormous energy loads. A dark asphalt parking lot absorbs approximately 850–950 W/m², driving surface temperatures to 150–170°F.

Thermal Mass Influence

- Concrete and masonry structures (common in commercial and industrial zones) have high thermal mass (specific heat ~0.88 J/g·K for concrete), meaning they absorb large quantities of heat during the day and re-radiate it slowly through the evening and night.
- This thermal lag effect is a primary driver of the **nighttime urban heat island** in San José. On a day reaching 105.3°F, stored thermal energy in concrete structures and asphalt surfaces will sustain elevated nighttime temperatures, potentially keeping overnight lows above 75–80°F in dense areas — preventing physiological recovery for residents without air conditioning.
- Tilt-up concrete buildings in North San José's industrial zones are particularly effective thermal batteries due to their large exposed wall areas and dark-colored roofs.

Glass Curtain Wall Effects

- Downtown San José's newer office towers (e.g., buildings along North First Street and near the Diridon Station area) feature extensive glass facades that create solar reflection "hot spots" on adjacent streets and buildings.
- Reflected solar radiation from glass facades can increase pedestrian-level radiant heat load by 50–150 W/m² in affected zones, significantly worsening thermal comfort at street level.
- Inside these buildings, the greenhouse effect through glazing increases cooling loads, which in turn increases HVAC waste heat rejection outdoors.

Heat Mitigation Techniques

- **California Title 24 (2022 code cycle):** Requires cool roofs on new and re-roofed commercial buildings (minimum aged solar reflectance of 0.63 for low-slope roofs). This is actively reducing the thermal footprint of new construction.
- **CALGreen (California Green Building Standards Code):** Mandates various sustainability features in new construction, including provisions that indirectly reduce heat island effects.
- Permeable pavements are increasingly specified in new developments per San José's stormwater management requirements, though they remain a small fraction of total paved area.
- High-albedo coatings on existing roofs are incentivized but not mandated for existing buildings, limiting retrofit adoption.

Building Stock Age

- A significant portion of San José's residential stock was built during the post-WWII suburban boom (1950s–1970s), featuring minimal insulation (often R-11 walls, R-19 attic or less), single-pane windows, and aging HVAC systems with low efficiency (SEER 8–10 vs. modern SEER 15+).
- These older homes require substantially more energy to cool, generating more waste heat per unit of indoor comfort. At 105.3°F, many older HVAC systems may be unable to maintain comfortable indoor temperatures, creating both an energy crisis and a health crisis.
- Newer construction (post-2010) meets significantly higher insulation and efficiency standards, but represents a minority of total housing stock.

Sources:

- California Energy Commission Title 24 Building Energy Efficiency Standards
- CALGreen Building Standards Code
- Lawrence Berkeley National Laboratory Heat Island Group research on cool roofs and urban albedo
- ASHRAE Handbook data on building material thermal properties
- US Census Bureau American Housing Survey data on building age for San José MSA

5. Urban Expansion & Development Trends

Population Growth Trajectory

- San José's population is approximately 1.0 million (2024 estimate), making it the largest city in the Bay Area and the 12th largest in the US. Growth has been moderate at approximately 0.5–1.0% annually in recent years, driven by the tech economy.



- The city's General Plan envisions accommodating significant additional growth (targeting capacity for ~1.3 million residents by 2040), primarily through densification along transit corridors rather than outward expansion.
- Each additional resident adds to aggregate energy demand, HVAC load, vehicle miles traveled, and anthropogenic heat generation. Densification concentrates these effects spatially.

Construction Trends

- **Vertical expansion downtown:** San José has seen a surge in high-rise residential and mixed-use development proposals downtown, particularly around the planned **Google Downtown West / Diridon Station** mega-development. This project alone envisions approximately 7.3 million square feet of office space, 5,900 housing units, and 500,000 square feet of retail — a massive increase in built density.
- **North San José:** Continued development of tech campuses and data centers, with some conversion of industrial land to mixed-use residential.
- **Infill development:** ADU (Accessory Dwelling Unit) construction is increasing rapidly under California's permissive ADU laws, adding density to existing residential neighborhoods without corresponding green infrastructure additions.
- **Horizontal sprawl:** Largely constrained by the city's Urban Growth Boundary and surrounding agricultural/open space protections (e.g., Santa Clara Valley Agricultural Plan).

Net Change in Impervious Surface

- Estimated: Net impervious surface area is **increasing** at approximately 0.5–1.0% per year in the urbanized area, driven by infill development, parking structure construction, and conversion of remaining vacant lots.
- Some offset comes from green stormwater infrastructure requirements in new developments and parking lot tree planting mandates, but the net trend is toward increased imperviousness.
- The Google Downtown West project includes commitments to significant green space and tree planting, which could partially offset the heat impact of the development if fully implemented.

Climate-Adaptive Urban Design

- San José adopted a **Climate Smart San José** plan with goals including carbon neutrality, urban greening, and resilience — but specific heat action plan elements (cooling centers, heat emergency protocols) are still developing.
- California's **Extreme Heat Action Plan** (2022) provides a state-level framework that San José is beginning to integrate into local planning.
- Updated building codes (Title 24 2022 cycle) require improved thermal performance, but apply only to new construction and major renovations.
- The city's **Urban Village** planning framework aims to create walkable, transit-oriented nodes — these could reduce vehicle heat if they successfully reduce VMT, but increased density without adequate green infrastructure could worsen local heat islands.

Near-Term Heat Profile Projection (5–10 Years)

- The combination of continued densification (especially the Google Downtown West project), increasing data center capacity, growing population, and climate change trends (more frequent extreme heat events) will likely **intensify the urban heat island effect by 1–3°F** in central San José over the next decade.
- Mitigation from cool roofs, solar installations, and tree planting will partially offset this, but current trajectories suggest net warming of the urban microclimate.
- Extreme heat events like the current 105.3°F reading are projected to become 2–3 times more frequent by the 2030s under moderate emissions scenarios, making urban heat resilience an increasingly urgent priority.

Sources:

- City of San José Envision 2040 General Plan and Urban Village planning documents
- Google Downtown West Mixed-Use Plan Environmental Impact Report
- Climate Smart San José plan
- California Extreme Heat Action Plan (2022)
- Cal-Adapt climate projection tools for Santa Clara County
- US Census Bureau population estimates for San José

6. Pedestrian Infrastructure & Outdoor Thermal Comfort



Walkability Characterization

- Downtown San José has moderate walkability (Walk Score typically 70–85 for the central core), with a grid street pattern, sidewalks on most streets, and pedestrian-oriented development around San Pedro Square, the SoFA (South First Area) district, and near San José State University.
- Outside downtown, walkability drops significantly. Suburban neighborhoods are auto-oriented with wide arterials, limited crosswalks, and sidewalks that are often narrow, poorly maintained, or lacking shade.
- At 105.3°F, even moderate walking distances become hazardous. A 10-minute walk on an exposed sidewalk at this temperature can raise core body temperature to dangerous levels, particularly for elderly, young, or health-compromised individuals.

Shading Infrastructure

- **Downtown:** Some street tree coverage along major pedestrian streets (e.g., portions of South First Street, San Fernando Street near SJSU), but coverage is inconsistent. Few covered walkways or arcades exist — San José's downtown lacks the continuous shade structures found in some desert cities.
- **Transit stops:** VTA (Valley Transportation Authority) bus stops have variable shade coverage. Many stops along major arterials have only a pole-mounted sign with no shelter, leaving transit riders fully exposed. Light rail stations generally have platform canopies.
- **Suburban arterials:** Major roads like Story Road, Tully Road, King Road, and Alum Rock Avenue have minimal to no street tree shade, wide exposed sidewalks, and extensive adjacent parking lots that radiate heat onto pedestrians.
- Estimated: Less than 25% of San José's sidewalk network has adequate shade coverage (defined as >50% canopy or structural shade during midday hours).

Mean Radiant Temperature (MRT) Conditions

- At 105.3°F ambient air temperature with clear skies (typical for a Bay Area heat wave), **MRT in exposed urban areas likely exceeds 140–160°F**. MRT accounts for direct solar radiation, reflected radiation from buildings and pavement, and longwave radiation from hot surfaces.
- **Sky view factor (SVF):** Downtown San José has moderate SVF (~0.4–0.6) due to building heights of 50–200 feet, providing some canyon shading on east-west streets during morning and afternoon. However, north-south streets receive full midday sun exposure. Suburban areas have high SVF (~0.7–0.9), meaning nearly full sky exposure and maximum radiant heat load.
- Glass curtain wall reflections in downtown can create localized MRT spikes exceeding 170°F on adjacent sidewalks, making certain building frontages particularly uncomfortable.
- At these MRT levels, the Universal Thermal Climate Index (UTCI) would classify conditions as **"extreme heat stress" (UTCI > 46°C / 115°F)**, indicating danger of heat stroke for prolonged outdoor exposure.

Heat Exposure Risk

- **Critical risk factors at 105.3°F:** Radiant heat from asphalt and concrete surfaces (surface temperatures 140–170°F), limited shade, low wind speeds typical during Bay Area heat waves (offshore flow patterns that drive extreme heat also suppress sea breeze ventilation), and high HVAC exhaust heat at pedestrian level from ground-floor commercial units.
- Pedestrians on wide arterials in east San José face the highest heat exposure: combination of minimal shade, high-albedo-deficit surfaces, heavy traffic exhaust, and limited access to air-conditioned spaces.
- Outdoor workers (construction, landscaping, delivery) face occupational heat illness risk well above Cal/OSHA high-heat thresholds (which trigger additional protections above 95°F).

Cooling Interventions

- **Misting systems:** Not widely deployed in San José's public spaces. Some restaurants and private venues may use them, but no systematic public misting infrastructure exists.
- **Cooling centers:** The city activates cooling centers (libraries, community centers) during heat emergencies, but these are indoor refuges rather than outdoor comfort interventions.
- **Heat-reflective pavements:** Pilot projects have been discussed but not widely implemented. Some newer developments use lighter-colored concrete or pavers.
- **Passive cooling corridors:** The Guadalupe River corridor provides a modest natural ventilation channel through downtown, but wind speeds during offshore heat events are typically very low (2–5 mph), limiting its effectiveness.

Cycling Infrastructure



- San José has an expanding bike lane network (~300+ miles of bikeways of various classes), but most lanes are unshaded, on-street facilities adjacent to hot asphalt.
- At 105.3°F, cycling on exposed bike lanes creates significant heat stress risk. Radiant heat from adjacent vehicle lanes and pavement, combined with metabolic heat from exertion, can push body temperatures to dangerous levels.
- The Guadalupe River Trail and other off-street paths offer slightly better conditions where they pass through riparian vegetation, but shade coverage is inconsistent.

Notable Pedestrian Comfort Zones and Deficits

- **Relative comfort zones:** San Pedro Square area downtown (~0.3 miles north of coordinates) — mature trees and some restaurant shade structures provide modest relief. Plaza de César Chávez — trees and some shade. San José State University campus — mature trees on portions of the campus provide pedestrian shade.
- **Severe deficit zones:** East San José arterials (Story Road, Tully Road, King Road corridors, ~2–4 miles east) — wide, exposed, minimal shade, high traffic, and heat-vulnerable populations. North First Street corridor — wide arterial with tech campuses set back behind parking lots, minimal pedestrian shade. Monterey Road corridor south of downtown — auto-oriented commercial strip with extensive exposed pavement.
- At 105.3°F, **no outdoor location in San José provides adequate thermal comfort** for sustained activity. Even the best-shaded areas exceed heat stress thresholds. The analysis identifies relative differences, but absolute conditions are dangerous everywhere outdoors.

Sources:

- Walk Score data for San José neighborhoods
- VTA transit infrastructure documentation
- Cal/OSHA Heat Illness Prevention Standard (Title 8, Section 3395)
- Published UTCI and MRT research for California urban environments
- City of San José Better Bike Plan 2025
- EPA guidance on urban heat island mitigation and pedestrian thermal comfort

Events Analysis

1. Extreme Weather Events & Heat Event History

Heat-Related Events

- **Baseline temperature conversion:** The provided average temperature of 40.74°C converts to approximately **105.3°F**. This is used as the reference throughout this analysis.
- **Note on coordinates:** 37.29°N, 121.83°W corresponds to the San José, California metropolitan area in the Santa Clara Valley (Silicon Valley). The longitude is interpreted as West given the geographic context.
- **September 2022 heatwave:** One of the most extreme and well-documented heat events in California history struck the San José area from approximately September 1–9, 2022. San José reached approximately 109–110°F during the peak, shattering all-time records. The event was driven by an exceptionally strong and persistent upper-level ridge (heat dome) centered over the Great Basin and extending into California. Multiple consecutive days exceeded 105°F in interior valley locations. This event triggered a statewide Flex Alert and grid emergency declarations from CAISO (California Independent System Operator) due to extreme electricity demand.
- **July 2006 California heatwave:** A prolonged, deadly heatwave struck the state from approximately July 15–26, 2006. San José and the Santa Clara Valley experienced sustained temperatures above 100°F for over a week. This event is associated with an estimated 600+ excess deaths statewide (documented in California Department of Public Health analyses). Nighttime temperatures remained dangerously elevated, preventing physiological recovery.
- **June 2021 Western heat dome:** While the most extreme impacts were in the Pacific Northwest, San José experienced temperatures around 105°F during this late June event, which was part of a broader omega blocking pattern.
- **Synoptic conditions preceding extreme heat:** Extreme heat in the Santa Clara Valley is typically driven by: (a) persistent upper-level high-pressure ridging (500 mb heights $\geq$ 594 dam), (b) offshore or downslope flow compressing and warming air as it descends from the Diablo Range to the east or the Santa Cruz Mountains to the west, (c) prolonged dry antecedent conditions depleting soil moisture and reducing evapotranspiration, (d) absence of the typical marine layer and sea breeze intrusion from the San Francisco Bay that normally moderates afternoon temperatures.
- **Seasonal heat risk window:** The primary extreme heat season for San José runs from approximately **mid-June through mid-October**, with peak risk in July–September. October 2 falls in the **late tail of the heat season**, a period when late-season heat events (often called "Indian summer" or "October heat spikes") can still occur, particularly when Santa Ana-type or Diablo wind patterns develop. Historically, early October heat events are less common but not unprecedented — they tend to be shorter in duration (2–4 days) but can be intense.



- **Government/institutional responses:** During the September 2022 event, California declared a grid emergency (Stage 2 and approaching Stage 3), issued statewide conservation alerts, and activated cooling centers across Santa Clara County. The City of San José opened multiple cooling centers in libraries and community centers. Cal/OSHA enforced outdoor heat illness prevention standards for workers.
- **Compound events:** The Santa Clara Valley has experienced multiple instances of compound heat + wildfire smoke events. The August–September 2020 period saw extreme heat coincide with massive wildfire smoke from the SCU Lightning Complex fires (which burned in the Diablo Range immediately east of San José), creating simultaneous thermal stress and hazardous air quality (AQI frequently exceeding 200). This compound exposure is particularly dangerous because it limits outdoor cooling strategies and exacerbates cardiovascular and respiratory distress.

Other Extreme Weather With Heat Relevance

- **Droughts:** The Santa Clara Valley is highly susceptible to multi-year droughts (2012–2016, 2020–2022 being major recent examples). Drought amplifies heat through the drought-heat feedback mechanism: depleted soil moisture shifts the surface energy balance toward sensible heat flux rather than latent heat (evapotranspiration), raising surface and air temperatures. During drought years, San José's summer and early fall temperatures tend to run 2–5°F above climatological normals. Reservoir levels in the Santa Clara Valley Water District system dropped critically during these periods, compounding water-use restrictions with heat stress.
- **Wildfires:** The Diablo Range and Santa Cruz Mountains flanking San José are fire-prone, particularly in late summer and early fall when vegetation is driest. Major fires include the SCU Lightning Complex (2020, ~400,000 acres), the Morgan Fire (2013), and numerous smaller fires. Wildfire smoke creates a paradoxical thermal effect: thick smoke can reduce daytime solar heating slightly but traps longwave radiation at night, elevating nighttime temperatures and preventing cooling. The primary health concern is the combined thermal + particulate matter (PM_{2.5}) exposure, which synergistically increases cardiovascular and respiratory emergency department visits.
- **Post-atmospheric river humidity:** Occasionally, early October atmospheric river events can precede or follow heat spikes. When heat follows heavy rain, saturated soils and elevated humidity can dramatically increase wet-bulb temperatures, shifting the health risk from dry heat to humid heat stress. Estimated: This pattern occurs roughly once every 3–5 years in early October for the San José area.

Sources:

- NWS San Francisco Bay Area (Weather Forecast Office) historical event summaries
- California Department of Public Health — 2006 heatwave mortality analysis
- CAISO grid emergency records
- CAL FIRE incident archives
- U.S. Drought Monitor historical records

2. Heatwave Frequency & Intensity

- **Heatwave definition for this region:** The National Weather Service (NWS) San Francisco Bay Area office issues an **Excessive Heat Warning** for the Santa Clara Valley when daytime temperatures are forecast to reach $\geq 105^{\circ}\text{F}$ and nighttime temperatures remain $\geq 75^{\circ}\text{F}$ for two or more consecutive days, or when any single day reaches $\geq 110^{\circ}\text{F}$. A **Heat Advisory** is issued when daytime highs reach $100\text{--}104^{\circ}\text{F}$ with warm overnight lows for an extended period. These thresholds are specific to the Bay Area's acclimatization profile.
- **Frequency:** San José experiences approximately **3–6 days per year** at or above 100°F based on recent climatology, and approximately **1–2 distinct heatwave events per year** meeting the multi-day criteria. Historically typical: Prior to 2000, San José averaged roughly 2–3 days per year above 100°F . The trend shows a clear increase, estimated at approximately **1–2 additional extreme heat days per decade** since the 1980s baseline.
- **Intensity metrics:**
 - Typical heatwave peak: $102\text{--}108^{\circ}\text{F}$
 - Record peak: approximately $109\text{--}110^{\circ}\text{F}$ (September 2022)
 - Typical duration: 2–5 days for significant events, with the most extreme events (2006, 2022) lasting 7–10 days
 - **Nighttime minimums:** During major heatwaves, overnight lows in San José frequently remain at $70\text{--}78^{\circ}\text{F}$, well above the health-recovery threshold of approximately 65°F . The urban core retains heat more effectively than surrounding areas due to the urban heat island effect, and the enclosed valley geography limits nocturnal drainage cooling.
- **Record-breaking events:** The September 2022 event stands as the most intense documented heatwave for San José, with the all-time record high of approximately 109°F recorded at San José International Airport (SJC). The July 2006 event, while slightly less extreme in peak temperature locally ($\sim 108^{\circ}\text{F}$), was more lethal statewide due to its prolonged duration and widespread geographic scope.
- **Trend context:** Heatwaves in the Santa Clara Valley are increasing in frequency, intensity, duration, and exhibiting earlier seasonal onset and later seasonal persistence. Attribution studies for California broadly indicate that anthropogenic



climate change has made events like the September 2022 heatwave significantly more likely (estimated 3–5× more probable than in a pre-industrial climate). The extension of heat risk into October is a notable emerging trend.

- **Classification of 105.3°F on October 2:**

- **Climate normal comparison:** The average high temperature for San José on October 2 is approximately **78–80°F**. The provided temperature of 105.3°F is approximately **25–27°F above the climatological normal** — an extreme departure.
- **Heatwave threshold comparison:** 105.3°F **meets or exceeds the NWS Excessive Heat Warning threshold** ($\geq 105^\circ\text{F}$ daytime). If nighttime temperatures are also elevated ($\geq 75^\circ\text{F}$), this would trigger the highest-tier heat alert.
- **Health-risk classification:** This temperature falls squarely in the **"Extreme Danger"** category on the NWS heat index scale (assuming even moderate humidity). **Explicit classification: EXCEEDING HEATWAVE THRESHOLD — extreme anomaly for the date.**

- **Heat dome and blocking pattern susceptibility:** The Santa Clara Valley is highly susceptible to heat dome events. The region sits in a zone where upper-level ridging over the Great Basin or eastern Pacific can effectively suppress the marine layer and onshore flow that normally moderates temperatures. The valley's topographic enclosure (Santa Cruz Mountains to the west/southwest, Diablo Range to the east) traps heated air and limits ventilation. Blocking patterns (omega blocks) centered near 130–140°W longitude are particularly effective at producing multi-day extreme heat in this location.

Sources:

- NWS San Francisco Bay Area — heat product criteria and climatological records
- NOAA/NCEI climate normals (1991–2020) for San José, CA (Station: SJC)
- Published climate attribution studies for Western U.S. heat events (e.g., World Weather Attribution consortium analyses)
- WRCC (Western Regional Climate Center) historical temperature records

3. Public Health Impacts & Heat Vulnerability

- **Heat-related health burden:** Santa Clara County has documented significant increases in emergency department visits for heat-related illness (heat exhaustion, heatstroke, dehydration, acute renal failure) during heatwave events. During the September 2022 event, California statewide saw a substantial surge in heat-related ED visits and deaths; Santa Clara County was among the affected jurisdictions. The July 2006 heatwave was associated with an estimated 600+ excess deaths statewide, with the Central Valley and inland Bay Area valleys disproportionately affected. Historically typical: Heat-related ED visits in Santa Clara County increase by an estimated 2–5× during major heatwave events compared to baseline.
- **Vulnerable populations:**
 - **Elderly (65+):** San José has a growing elderly population; those living alone without air conditioning are at highest risk. Many older homes in San José were built without central AC, reflecting the historically mild climate.
 - **Outdoor workers:** The Santa Clara Valley has a significant agricultural workforce (southern San José, Coyote Valley) and construction workforce, both highly exposed to occupational heat illness.
 - **Unhoused populations:** San José has one of the largest unhoused populations in the United States (estimated 6,000–10,000 individuals in Santa Clara County), with major encampments along Coyote Creek and in downtown areas with minimal shade or cooling access.
 - **Low-income communities:** East San José (ZIP codes 95116, 95122, 95127) and parts of central San José have higher proportions of low-income residents, older housing stock without AC, and less tree canopy — creating concentrated heat vulnerability.
 - **Children:** Particularly at risk during school activities and outdoor play; early October is within the school year, increasing exposure during recess and outdoor PE.
- **Heat-mortality relationship:** Estimated: For the San José/Bay Area region, excess mortality begins to increase measurably when daily maximum temperatures exceed approximately **95–100°F**, reflecting the population's relatively low acclimatization to extreme heat compared to desert Southwest cities. At 105.3°F, the population is well into the elevated mortality risk zone. The mortality risk curve steepens sharply above 100°F for Bay Area populations because of lower baseline AC prevalence and lower physiological acclimatization.
- **Consecutive hot night risk:** At an average temperature of 105.3°F, nighttime minimums are likely to remain at or above 75–80°F, well above the critical **75°F (24°C)** threshold for physiological recovery. Sustained nighttime heat prevents the body from dissipating accumulated thermal load, leading to cumulative heat stress. This is particularly dangerous for elderly individuals and those without air conditioning. If this temperature represents a multi-day event, the consecutive hot night effect would significantly compound mortality risk — studies show heat mortality roughly doubles after the second consecutive night above recovery thresholds.
- **Heat advisory and warning systems:**
 - The NWS issues tiered alerts: **Heat Advisory** (moderate risk), **Excessive Heat Watch** (potential for extreme heat), and **Excessive Heat Warning** (imminent or occurring extreme heat). At 105.3°F, an Excessive Heat Warning would almost certainly be in effect.



- Santa Clara County has an Operational Area Emergency Operations Center that activates during extreme heat events, coordinating with the County Public Health Department.
- The California Heat Assessment Tool (CHAT) provides localized heat-health risk assessments for planning purposes.
- **Cooling infrastructure and public health interventions:**
 - Santa Clara County activates **cooling centers** at libraries, community centers, and senior centers during heat events. The City of San José maintains a list of designated cooling locations.
 - **Cal/OSHA Heat Illness Prevention Standard** (Title 8, Section 3395) mandates shade, water, rest breaks, and high-heat procedures ($\geq 95^{\circ}\text{F}$) for outdoor workers — among the strongest occupational heat protections in the U.S.
 - California's CPUC (California Public Utilities Commission) has enacted **utility shutoff moratoriums** during extreme heat events, preventing disconnection of electricity for nonpayment during declared heat emergencies.
 - The City of San José has invested in urban tree canopy expansion and cool-pavement pilot programs, though coverage remains uneven across neighborhoods.
- **Heat vulnerability mapping:** Santa Clara County Public Health Department has published heat vulnerability assessments identifying East San José, downtown San José, and parts of Gilroy/Morgan Hill as highest-risk areas. The CalEPA CalEnviroScreen tool also identifies census tracts in East San José (near the coordinates provided) as scoring high on pollution burden and population vulnerability indicators. The Trust for Public Land and Climate Central have also produced heat-island mapping showing significant intra-urban temperature variation of $10\text{--}15^{\circ}\text{F}$ between well-treed neighborhoods (e.g., Willow Glen, Rose Garden) and industrial/commercial zones (e.g., North San José, Alum Rock).
- **Health-system preparedness:** Historically typical: During major heat events, Santa Clara Valley Medical Center (the county's safety-net hospital) and regional hospitals experience surge in heat-related admissions. EMS call volumes increase by an estimated $15\text{--}30\%$ during multi-day heatwaves. The September 2022 event strained ambulance response times in parts of the county. The county's Emergency Medical Services Agency has heat-specific protocols for paramedic response.
- **Equity dimension:** Heat health burden in San José is disproportionately concentrated in **East San José and the urban core**, areas with higher proportions of Latino/Hispanic and Vietnamese American residents, lower median incomes, less tree canopy (as low as $5\text{--}8\%$ coverage vs. $20\text{--}25\%$ in wealthier neighborhoods), higher impervious surface area, and lower rates of residential air conditioning. These neighborhoods can be $8\text{--}15^{\circ}\text{F}$ hotter than adjacent affluent areas during peak heat. This creates a stark environmental justice dimension where communities of color and low-income residents bear the greatest heat health burden despite contributing least to the emissions driving climate change.

Sources:

- Santa Clara County Public Health Department — heat vulnerability assessments and emergency preparedness plans
- California Department of Public Health — heat-related morbidity and mortality surveillance data
- Cal/OSHA Heat Illness Prevention Standard (Title 8, CCR §3395)
- CalEPA CalEnviroScreen 4.0
- California Heat Assessment Tool (CHAT)
- Trust for Public Land and Climate Central urban heat island analyses
- NWS San Francisco Bay Area — heat alert criteria and historical warnings

Anthropogenic Factors Analysis

1. Heat Emissions from Vehicles & Industry

Temperature Context

- The provided average temperature of 40.74°C converts to **105.3°F** — an exceptionally high reading for San José, California (the coordinates correspond to the San José metropolitan area in Santa Clara County). This temperature is well above the city's typical early-October average high of approximately $78\text{--}82^{\circ}\text{F}$, suggesting an extreme heat event or localized anomaly.

Transportation Heat Footprint

- San José and the broader Silicon Valley region are heavily **car-dependent**. The dominant transport mode for commuting is the single-occupancy vehicle, with approximately $75\text{--}80\%$ of commuters driving alone based on Census Bureau American Community Survey data.
- Major vehicular waste heat corridors include **US-101, I-280, I-680, and CA-87 (Guadalupe Freeway)** all of which carry high traffic volumes through the urban core. US-101 through San José regularly handles over 200,000 average daily trips in its busiest segments.
- Vehicular waste heat is generated through engine combustion exhaust ($\sim 60\%$ of fuel energy is lost as heat), tire-road



friction (adding localized surface heating of 2–5°F above ambient pavement temperatures), and brake heat at congested intersections and freeway on/off-ramps.

- At 12:00 hours on a Wednesday (October 2, 2024), traffic is in a **midday lull between morning and evening peaks**, but commercial vehicle traffic, delivery trucks, and ride-hail services maintain significant heat output. Lunchtime traffic in downtown San José and commercial corridors (Santana Row, Valley Fair area, North First Street) generates concentrated waste heat.
- **Estimated:** Highway corridors in similar California metro areas typically exhibit surface temperature differentials of 5–12°F above surrounding vegetated areas, compounded by vehicle exhaust at this extreme ambient temperature.

Industrial and Commercial Waste Heat

- Silicon Valley is home to an exceptionally high concentration of **data centers**, which are among the most significant point sources of anthropogenic heat in the region. Major facilities operated by Equinix, Vantage Data Centers, CyrusOne, and others are clustered in north San José (particularly along North First Street and near Mineta San José International Airport).
- **Estimated:** A single large data center (20+ MW IT load) rejects approximately 60–70 million BTU/hour of waste heat to the atmosphere via cooling towers and condenser systems. The north San José corridor likely hosts aggregate data center capacity exceeding 200 MW, making this one of the densest data center heat rejection zones in North America.
- The **San José International Airport (SJC)** contributes waste heat through jet engine exhaust, ground support equipment, and extensive asphalt apron surfaces. Aircraft taxiing and idling can produce localized temperature increases of 3–7°F on apron surfaces.
- Traditional heavy industry is relatively limited in San José compared to manufacturing-heavy cities. The region's industrial heat profile is dominated by **semiconductor fabrication facilities** (fabs), electronics manufacturing, and commercial food service operations rather than refineries or steel mills.
- The former Spartan Keyes and Monterey Road industrial corridors contain light manufacturing, warehousing, and distribution operations that contribute modest waste heat.

Temporal Patterns at the Given Datetime

- At **12:00 hours on a Wednesday**, commercial and office HVAC systems are at or near peak operation, data centers are running at full capacity (they operate 24/7 but cooling loads peak during afternoon heat), and midday delivery and commercial vehicle traffic is active.
- Industrial shift schedules in the region are predominantly daytime (7 AM–6 PM), so waste heat from manufacturing and fab operations is near peak output.
- Data centers represent a **constant baseload heat source** — their waste heat output varies less by time of day than by seasonal cooling demand, and at 105.3°F, their cooling systems are working at maximum capacity, rejecting more heat per unit of IT load than under mild conditions.

Emission-Reduction Measures

- California leads the nation in EV adoption. Santa Clara County has one of the highest EV registration rates in the U.S., with battery electric vehicles comprising an estimated 20–25% of new vehicle sales as of 2024. EVs produce significantly less waste heat per mile than internal combustion vehicles (no exhaust heat, higher drivetrain efficiency), though tire-road friction and brake heat persist.
- San José has committed to a **Climate Smart San José** plan targeting carbon neutrality, which includes fleet electrification for municipal vehicles and expansion of EV charging infrastructure.
- VTA (Valley Transportation Authority) has been transitioning to electric buses, reducing per-vehicle waste heat from transit operations.
- Several data center operators in the region are exploring or implementing **waste heat recovery** for adjacent commercial heating needs, though widespread district heating from data center waste heat remains limited due to the mild climate's low heating demand.

Waste Heat Recovery Opportunities

- The concentration of data centers in north San José represents a significant untapped waste heat resource. **Estimated:** At typical data center exhaust temperatures of 95–115°F, this heat is suitable for domestic hot water preheating, swimming pool heating, or absorption chiller input, but district heating demand in San José's mild climate is low.
- Some semiconductor fabs reclaim process heat internally for facility heating and process water preheating.
- The San José-Santa Clara Regional Wastewater Facility processes significant volumes and could theoretically integrate waste heat from nearby industrial sources, though no such program is widely documented.



Sources:

- U.S. Census Bureau American Community Survey — commute mode share data
- Caltrans Traffic Census Program — highway volume data
- Climate Smart San José plan (City of San José)
- California Energy Commission — EV adoption statistics
- General knowledge of Silicon Valley data center geography and thermal engineering principles

2. Cooling Infrastructure & Waste Heat Feedback Loop

Cooling Infrastructure Prevalence

- San José has a historically **moderate AC penetration rate** compared to hotter California metros like Fresno or Sacramento. Estimated residential AC penetration is approximately 60–70%, which is below the national average of ~90% and significantly below inland California cities (85–95%). This reflects San José's typically mild Mediterranean climate where many older homes were built without central air conditioning.
- Commercial and institutional buildings have near-universal AC coverage (~95%+), as modern building codes and tenant expectations require it.
- The relatively lower residential AC penetration becomes a critical vulnerability during extreme heat events like the one indicated by 105.3°F.

The AC Feedback Loop

- At 105.3°F, virtually every air conditioning system in the metro area is operating at or near maximum capacity. Each unit rejects approximately **1.2–1.3 units of heat outdoors for every unit of cooling delivered indoors** (based on a COP of approximately 2.5–3.5 at this extreme temperature, degraded from rated performance).
- AC condenser efficiency drops significantly at ambient temperatures above 100°F. **Estimated:** COP degradation of 2–4% per °F above 95°F means systems consume more electricity and reject proportionally more waste heat at 105.3°F — a vicious feedback loop.
- In dense commercial zones (downtown San José, Santana Row, North First Street office corridor), concentrated condenser waste heat can elevate local ambient temperatures by an **estimated 1–3°F**, particularly in street canyons and areas with rooftop condenser clusters.
- Nighttime persistence of this effect is significant: condenser waste heat continues into evening hours as buildings attempt to cool down thermal mass, contributing to elevated overnight minimum temperatures.

Balance-Point Temperature

- The typical cooling balance-point temperature for this climate zone (ASHRAE Climate Zone 3C, warm-dry) is approximately **65°F**, the threshold above which buildings begin requiring mechanical cooling.
- At 105.3°F, the current temperature is **40.3°F above the balance point** — an extreme deviation that places cooling systems under maximum stress.
- Cooling Degree Days (CDD) for this single day: using a 65°F base, the CDD contribution is approximately $(105.3 - 65) / 2 \approx 20$ CDD for this day alone (simplified estimate), which is extraordinary for San José where the annual CDD total is typically around 600–900.

Cooling Inequality

- Significant disparities exist in San José. **East San José** (neighborhoods like Alum Rock, Mayfair) has older housing stock, lower household incomes, and lower AC penetration rates. Many homes rely on window units or evaporative coolers, which are ineffective at 105.3°F with low humidity.
- Rental housing, which comprises a large share of San José's housing stock (~55–60% of units), often lacks central AC, and tenants may face landlord resistance to installation or inability to afford electricity costs for portable units.
- Mobile home parks (several exist in south and east San José) are particularly vulnerable — poor insulation and limited cooling capacity make them dangerous during extreme heat.
- The City of San José activates **cooling centers** during extreme heat events, but access depends on transportation and awareness, creating gaps for elderly, disabled, and non-English-speaking populations.

District Cooling Systems

- San José does not have a significant public district cooling system. Cooling is predominantly handled by individual



building systems.

- Some large campus developments (e.g., major tech company campuses in north San José and adjacent areas) operate centralized chilled water plants that are more efficient than distributed rooftop units, but these serve private campuses rather than the public.
- The planned **Google Downtown West** mixed-use development includes district-scale energy systems that may incorporate centralized cooling, representing a future efficiency improvement.

Innovative and Passive Cooling Strategies

- San José's Climate Smart plan promotes **cool roofs and cool pavements**, which reduce cooling demand by lowering surface temperatures, though adoption rates remain modest.
- The region's dry climate makes **nighttime ventilation flushing** effective during normal conditions (large diurnal temperature swings), but at 105.3°F daytime temperatures, nighttime lows may remain too high for effective passive cooling.
- Evaporative cooling is partially effective given the region's low humidity, but at extreme temperatures its capacity is limited.
- Some newer commercial buildings incorporate **radiant cooling** and ground-source heat pump systems, which maintain higher efficiency at extreme ambient temperatures compared to air-cooled condensers.

Cooling Demand Classification

- At 105.3°F in San José — a city whose infrastructure was designed for a much milder climate — cooling demand is classified as **EMERGENCY**. This temperature exceeds the city's design-day conditions, pushes AC systems beyond rated capacity, triggers public health cooling advisories, and creates dangerous conditions for the significant population segment without adequate cooling access.

Sources:

- U.S. Energy Information Administration — Residential Energy Consumption Survey (RECS) for AC penetration
- ASHRAE climate zone classifications
- City of San José Climate Smart plan and heat emergency protocols
- General HVAC engineering principles for COP degradation at high ambient temperatures

3. Energy System Load & Grid Stress

Energy Demand Profile

- San José is served primarily by **Pacific Gas & Electric (PG&E)** for electricity and natural gas, with **San José Clean Energy (SJCE)** serving as the community choice energy provider (procuring electricity supply while PG&E maintains distribution).
- The consumption split is approximately: **commercial/tech sector ~50–55%** (driven heavily by data centers and office campuses), **residential ~30–35%**, and **industrial ~10–15%**. The tech sector's dominance in electricity consumption is a defining feature of this area.
- San José Clean Energy sources approximately 60%+ of its electricity from carbon-free sources, including solar, wind, and large hydro. California's overall grid mix includes significant solar (peaking midday), wind, natural gas peaker plants, nuclear (Diablo Canyon), and imported hydropower.
- Natural gas remains significant for residential water heating and cooking, and critically, for **gas-fired peaker plants** that activate during extreme demand — these plants themselves generate substantial waste heat.

Temperature-Demand Sensitivity

- The California Independent System Operator (CAISO) has documented that statewide electricity demand increases approximately **800–1,000 MW per °F** above ~85°F during heat events. At 105.3°F, this represents roughly 20,000+ MW of additional cooling-driven demand above baseline.
- For the San José area specifically, **Estimated**: local demand may increase 3–5% per °F above 80°F, driven primarily by residential and commercial AC load. At 105.3°F, this implies local demand could be 75–125% above mild-weather baseline.
- The CDD baseline for CAISO planning is 65°F. At 105.3°F, the cooling degree intensity is extreme and would place this day among the highest-demand days of the year.

Grid Stability and Heat-Related Risk



- California has a well-documented history of heat-related grid stress. The most notable recent events include the **August 2020 rolling blackouts** (CAISO ordered Stage 3 emergencies during a multi-day heat wave, resulting in the first rolling blackouts since 2001) and the **September 2022 heat dome** where CAISO issued Flex Alerts and came within minutes of ordering blackouts as demand hit a record ~52,000 MW.
- At 105.3°F in early October, this would represent an **unusually late-season extreme heat event**. Grid risk is compounded by the fact that some generation resources may already be in planned maintenance for the fall shoulder season, and solar production begins declining in October (shorter days, lower solar angle).
- PG&E distribution infrastructure in San José faces **heat derating**: underground cables and pad-mounted transformers lose capacity at sustained high temperatures. Transformer failure risk increases significantly above 100°F ambient, particularly in older neighborhoods with aging infrastructure.
- Transmission lines experience increased resistive losses and thermal sag at extreme temperatures, reducing effective transfer capacity by an estimated 2–5% at these temperatures.

Grid Heat Vulnerability

- The concentration of data centers in north San José creates localized **extreme load density**. A cluster of data centers drawing 200+ MW from a limited number of substations creates single-point-of-failure risks during heat events when transformer capacity is derated.
- Residential neighborhoods in east and south San José with older distribution infrastructure are most vulnerable to localized outages during extreme heat.

Load Shedding Risk

- This is **highly relevant** for this location. California has an active and recent history of load shedding during heat events. At 105.3°F, CAISO would almost certainly issue **Flex Alerts** calling for voluntary conservation, and depending on the geographic extent of the heat event, could escalate to Stage 1–3 grid emergencies.
- PG&E's Public Safety Power Shutoffs (PSPS) are primarily wildfire-related rather than heat-related, but extreme heat combined with dry conditions could trigger both grid stress and fire risk simultaneously.

Energy Resilience Measures

- California leads the nation in **battery energy storage deployment**. As of 2024, CAISO has approximately 10,000+ MW of grid-scale battery storage, much of it deployed since 2020, which significantly improves the grid's ability to manage evening ramp periods when solar drops off but cooling demand persists.
- San José Clean Energy participates in **demand response programs** that can curtail commercial and industrial load during emergencies.
- Rooftop solar penetration in San José is substantial (Santa Clara County has among the highest residential solar adoption rates in the state), providing distributed generation that reduces grid stress during peak solar hours — though at 12:00 noon, solar is near peak output, which partially offsets cooling demand.
- At the given time (12:00 noon), solar generation is near its daily maximum, providing a buffer. The **critical stress period** would be 4:00–8:00 PM when solar production drops rapidly but cooling demand remains extreme — the "net demand peak" or "duck curve" belly.

Energy System Demand Curve Assessment

- At 105.3°F and noon, the energy system is operating at **near-peak demand**, partially mitigated by strong midday solar generation. The system is likely within 85–95% of its daily peak, which will occur in late afternoon. Grid stress is **high and escalating**, with the most dangerous period still 4–6 hours away.

Sources:

- California Independent System Operator (CAISO) — grid operations data and emergency history
- San José Clean Energy — energy procurement and program information
- PG&E service territory infrastructure knowledge
- California Energy Commission — battery storage deployment data
- CAISO 2020 rolling blackout root cause analysis report

4. Traffic Behavior & Vehicular Heat Contribution



Traffic Pattern Characterization

- San José is a **car-dependent metro** with a sprawling land use pattern. The city consistently ranks among the most congested in California, though less so than Los Angeles or the San Francisco core.
- Peak commute hours are approximately 7:00–9:30 AM and 4:30–7:00 PM on weekdays. At 12:00 noon on a Wednesday, traffic is in a **midday inter-peak period** — lower than rush hour but still significant due to commercial activity, deliveries, and tech campus shuttle movements.
- The dominant commute pattern involves north-south flows along US-101 and I-880 (connecting San José to other Silicon Valley employment centers and San Francisco), and east-west flows on I-280 and local arterials.

Waste Heat Concentration Zones

- **US-101/I-680/I-280 interchange complex:** One of the busiest interchange systems in the South Bay, with concentrated stop-and-go traffic generating intense localized waste heat from idling engines and brake application.
- **North First Street corridor:** Heavy traffic serving data centers, tech campuses, and the airport creates a combined vehicle + building waste heat zone.
- **Mineta San José International Airport:** Aircraft operations, ground vehicles, and extensive paved surfaces create a significant heat island. Jet blast from departing aircraft can raise local temperatures by 10–20°F in the immediate vicinity.
- **Downtown San José:** Concentrated parking structures (multiple multi-story garages), ride-hail pickup/dropoff zones, and bus transit corridors create localized heat pockets.
- **Freight corridors:** Oakland Road and other routes connecting to industrial/warehouse districts in north and east San José carry heavy truck traffic that generates disproportionate waste heat per vehicle.
- **Estimated:** Major freeway corridors in the Bay Area exhibit surface temperature differentials of 8–15°F above adjacent vegetated areas during extreme heat events, compounded by vehicle waste heat contributions.

Pavement Heat Interaction

- At 105.3°F ambient temperature, asphalt pavement surface temperatures on San José roadways likely reach **140–160°F**. Vehicle traffic prevents radiative cooling by maintaining surface agitation and depositing additional heat through tire friction and exhaust.
- Tire-road friction adds approximately 0.5–1.0°F of surface temperature per 1,000 vehicles per hour on a given road segment. On US-101 segments carrying 10,000+ vehicles/hour during peaks, this is a measurable contribution.
- Dark asphalt (albedo ~0.05–0.10) absorbs 90–95% of incoming solar radiation. San José has begun limited deployment of **cool pavement coatings** (higher albedo sealants) on some streets, but coverage remains minimal.

Public Transit Role

- VTA operates light rail and bus service, and Caltrain provides commuter rail. However, transit mode share in San José is low — approximately **3–5% of commute trips**, far below San Francisco's ~30%+.
- The low transit share means the vast majority of person-trips generate individual vehicle waste heat rather than being consolidated into more efficient mass transit.
- VTA's light rail system is electrically powered, producing no direct exhaust heat (though electricity generation creates upstream heat). Diesel buses, still in partial service, contribute exhaust heat directly to street-level air.
- BART's extension to San José (completed 2024) may modestly shift some commuters from cars to electric rail, reducing per-trip waste heat generation.

Heat-Induced Behavioral Shifts

- At 105.3°F, significant behavioral changes are expected:
 - Walking and cycling trips are sharply reduced — pedestrians and cyclists face dangerous heat exposure.
 - Transit ridership may decline due to exposed waiting areas (many VTA stops lack adequate shade), pushing riders to air-conditioned private vehicles.
 - Drive-through restaurant and coffee shop queues lengthen as people avoid exiting vehicles, creating idling vehicle heat clusters.
 - Some outdoor workers and construction crews may be on reduced schedules or heat breaks per Cal/OSHA requirements, shifting some vehicle trips to earlier or later hours.

Mitigation Measures



- San José does not currently operate congestion pricing or formal low-emission zones.
- VTA is progressively deploying **electric buses**, with a goal of full fleet electrification aligned with California's Innovative Clean Transit regulation (full zero-emission fleet by 2040).
- The city has invested in **protected bike lanes** and pedestrian improvements, though usage drops dramatically during extreme heat events.
- Smart traffic signal coordination on major arterials (e.g., adaptive signal control on Stevens Creek Boulevard, El Camino Real) reduces stop-and-go idling, modestly decreasing waste heat from braking and acceleration cycles.
- Limited shaded transit stop improvements have been implemented, but coverage is inconsistent.

Datetime Traffic Assessment

- At 12:00 noon on Wednesday, October 2, 2024, traffic is in a **moderate midday pattern** — approximately 60–70% of peak-hour volume. Waste heat generation from vehicles is significant but not at its daily maximum. The combination of midday traffic heat and extreme ambient temperature creates compounding thermal stress on roadway corridors. The afternoon rush hour (beginning ~4:00 PM) will bring the day's peak vehicular waste heat load.

Sources:

- Caltrans Performance Measurement System (PeMS) — traffic volume characterization
- VTA transit planning documents
- Cal/OSHA heat illness prevention standards
- City of San José transportation planning documents
- General pavement thermal engineering literature

5. Seasonal Population Dynamics & Heat Demand

Population Variability

- San José does **not experience dramatic seasonal population swings** comparable to tourist-driven or university-dominated cities. The population is approximately 1 million residents with a relatively stable year-round base.
- However, several secondary population dynamics are relevant:
 - **San José State University (SJSU)** and several community colleges are in full fall session as of October 2, adding approximately 35,000–40,000 students to the downtown and surrounding areas. This is a **high-population period** compared to summer break.
 - The tech workforce has partially shifted to hybrid work patterns post-pandemic, meaning weekday office occupancy in Silicon Valley is typically 60–80% of pre-2020 levels. This moderately reduces commercial building cooling loads compared to full occupancy.
 - San José hosts periodic large events (San José Sharks hockey season begins in October, conventions at the San José Convention Center), which create localized population surges downtown.

Demand Translation

- The return of university students in fall increases residential energy demand in the SJSU area (dormitories, nearby apartments), increases transit and vehicle traffic, and adds cooling load to campus buildings.
- Full weekday commercial occupancy (Wednesday) means office buildings, retail, and restaurants are operating at typical capacity, driving commercial cooling demand to its weekly maximum.
- Hybrid work patterns mean some residential areas see higher midday cooling demand than pre-pandemic (more people working from home), while some commercial buildings operate at reduced but still significant occupancy.

Current Season Assessment

- Early October in San José is typically the **transition from warm-dry summer to the beginning of the cool-wet season**. Normal temperatures would be declining, and energy demand would be moderating from summer peaks.
- At 105.3°F, this represents a **severe anomaly** — the population and infrastructure are not expecting extreme heat in October. Key implications:
 - Some commercial HVAC systems may have been partially serviced or scheduled for maintenance during the expected shoulder season, reducing available cooling capacity.
 - Public awareness and behavioral adaptation to extreme heat may be lower than during a July/August heat wave, increasing health risk.
 - Energy system planning typically does not anticipate peak cooling demand in October, meaning reserve margins



may be tighter.

- The anthropogenic heat load is at a **high level** — full weekday commercial and institutional activity combined with emergency-level cooling demand creates a heat output profile comparable to or exceeding a typical peak summer day.

Weekend/Weekday and Time-of-Day Demand Variation

- **Wednesday at noon** represents near-peak weekly commercial energy demand. Office buildings, retail, data centers, and institutional facilities are all fully operational.
- Commercial electricity demand at this time is typically 30–40% higher than weekend levels for the same temperature.
- Residential demand at noon on a weekday is elevated above normal due to the extreme heat (people running AC even if away from home to protect pets, prevent heat damage to electronics/furnishings, or because they are working from home).
- Restaurant and food service operations are ramping up for lunch, adding commercial kitchen waste heat and additional HVAC load.

Infrastructure and Policy Adjustments

- The City of San José activates its **Extreme Heat Response Plan** during events like this, which includes opening cooling centers at libraries and community centers, extending pool hours, and conducting welfare checks on vulnerable populations.
- PG&E and CAISO issue **Flex Alerts** requesting voluntary conservation during afternoon peak hours (typically 4–9 PM).
- VTA may deploy additional vehicles on high-ridership routes to reduce wait times at exposed stops.
- San José does not have significant seasonal energy pricing variation, but time-of-use (TOU) rates under PG&E/SJCE tariffs charge higher rates during peak hours (4–9 PM), providing a price signal to shift demand.
- No significant temporary cooling infrastructure (such as misting stations or temporary shade structures) is routinely deployed in San José, though individual businesses and event venues may implement ad hoc measures.

Sources:

- San José State University enrollment data
- City of San José Emergency Services — Extreme Heat Response protocols
- CAISO Flex Alert program
- PG&E time-of-use rate schedules
- General knowledge of Silicon Valley workforce and demographic patterns

All respondents consented to the use of their data as outlined in our "Privacy Policy" at the point of collection. [Privacy Policy](#)